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Machine Learning Accelerated Multi-Agent Cross-Layer Exploration for Perovskite Solar Cell Additive Design: A Data-Driven Approach to Self-Assembled Monolayer and Molecular Modulator Screening

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Abstract

Perovskite solar cells (PSCs) have emerged as one of the most promising photovoltaic technologies, with power conversion efficiencies (PCEs) exceeding 27% in laboratory settings. A critical pathway to further efficiency gains lies in the rational design of molecular additives and self-assembled monolayers (SAMs) that modulate charge transport, passivate defects, and stabilize interfaces. However, traditional empirical trial-and-error approaches for molecular additive discovery are time-consuming, costly, and often fail to exploit the full combinatorial space of candidate molecules. Here, we present a **multi-agent parallel cross-layer exploration framework** that automates the discovery of optimal machine learning pipelines for PSC additive design. Our system simultaneously explores combinations across five layers—data cleaning, feature engineering, model selection, evaluation strategy, and virtual screening—using weighted random sampling and multiprocessing-based parallel execution. By systematically evaluating 12+ model-feature-evaluation combinations on a curated dataset of 4,934 PSC devices with molecular modulators, we discovered that **MACCS fingerprints combined with Random Forest and random-split evaluation** achieve the highest cross-validation R^2 of **0.296** for predicting ΔPCE (the PCE improvement induced by additives), surpassing our previous best of 0.284. Furthermore, we demonstrate that including the baseline PCE (without modulator) as an input feature improves prediction accuracy by **188%**, and that an agentic "VeryLoose"data cleaning strategy optimally balances sample retention with signal quality. SHAP interpretability analysis reveals that electrotopological state descriptors (EState_VSA), aromatic substructures, and lipophilicity distributions are the most influential molecular features governing additive performance. Based on model insights, we designed three novel SAM candidates with predicted PCEs approaching 27%. This work establishes a scalable, automated framework for accelerated molecular additive discovery in PSCs and provides actionable design rules for the photovoltaic community.

Keywords: perovskite solar cells, machine learning, self-assembled monolayers, molecular additives, multi-agent system, automated machine learning, virtual screening, SHAP interpretability

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1. Introduction

1.1 The Challenge of Molecular Additive Design in Perovskites

Metal halide perovskites have revolutionized the photovoltaic landscape over the past decade, with certified PCEs rising from 3.8% in 2009 to over 27% today.[1] This remarkable progress stems from intensive research into perovskite composition engineering, interface optimization, and device architecture refinement. Among the various strategies for efficiency enhancement, molecular additives—including self-assembled monolayers (SAMs), passivating agents, hole transport materials (HTMs), and electron transport materials (ETMs)—have proven indispensable for defect passivation, energy level alignment, and moisture protection.[2]

SAMs, in particular, have attracted significant attention as hole-selective contacts in inverted (p-i-n) PSCs due to their minimal parasitic absorption, low material consumption, and straightforward solution-processing compatibility.[3] Molecules such as MeO-2PACz and related phosphonic acid-anchored SAMs have enabled PCEs exceeding 25% by forming ordered monolayers that promote efficient hole extraction while blocking electrons.[4] Similarly, molecular modulators added to the perovskite precursor solution can control crystallization kinetics, reduce trap state densities, and enhance phase stability.[5]

Despite these successes, the discovery of high-performance additives remains predominantly empirical. Researchers typically synthesize and test dozens of candidate molecules—a process that requires substantial synthetic effort, device fabrication expertise, and characterization time. The combinatorial space of potential additives is vast: PubChem alone contains over 110 million molecules, and even within plausible ranges of molecular weight, lipophilicity, and functional groups, millions of candidates could theoretically serve as PSC additives.[6] This mismatch between the enormous search space and the limited throughput of experimental screening creates a compelling need for computational acceleration.

1.2 Machine Learning in Perovskite Materials Discovery

Machine learning (ML) has emerged as a powerful tool for structure-property relationship modeling in materials science. In the perovskite domain, ML has been successfully applied to predict band gaps, formation energies, ionic conductivities, and device PCEs.[7] Notably, Yang et al. (2025) demonstrated an R^2 of 0.76 for PCE prediction using 2,079 experimental devices, validated by 12 independent experiments with an average error of 1.6%.[8] Li et al. (2026) recently reported a dedicated ML platform for SAM design in inverted PSCs, achieving an R^2 of 0.51 with RDKit descriptors and XGBoost, validated against newly synthesized molecules.[9]

These studies, however, share a common limitation: each selects a single pipeline (one data source, one feature set, one model, one evaluation strategy) and pursues it to completion. The cross-layer combinatorial space—spanning data cleaning methods, molecular

[1] Green, M. A.; Dunlop, E. D.; Yoshita, M.; Kopidakis, N.; Bothe, K.; Siefer, G. Solar Cell Efficiency Tables (Version 63). Prog. Photovoltaics Res. Appl. **2024**, 32, 583–593.

[2] Park, N.-G. Research Update: Long-Term Stability of Perovskite Solar Cells—Novel Assessment and Perspectives. APL Mater. **2024**, 12, 010901.

[3] Jeong, J.; Kim, M.; Seo, J.; Lu, H.; Ahlawat, P.; Mishra, A.; Yang, Y.; Hope, M. A.; Eickemeyer, F. T.; Kim, M.; et al. Pseudo-Halide Anion Engineering for α-$FAPbI_3$ Perovskite Solar Cells. Nature **2021**, 592, 381–385.

[4] Aydin, E.; De Bastiani, M.; De Wolf, S. Defect and Contact Passivation for Perovskite Solar Cells. Adv. Mater. **2019**, 31, 1900428.

[5] Li, Z.; Klein, T. R.; Kim, D. H.; Yang, M.; Berry, J. J.; van Hest, M. F.; Zhu, K. Scalable Fabrication of Perovskite Solar Cells. Nat. Rev. Mater. **2018**, 3, 18017.

[6] Kim, S.; Chen, J.; Cheng, T.; Gindulyte, A.; He, J.; He, S.; Li, Q.; Shoemaker, B. A.; Thiessen, P. A.; Yu, B.; et al. PubChem 2023 Update. Nucleic Acids Res. **2023**, 51, D1373–D1380.

[7] Pilania, G.; Wang, C.; Jiang, X.; Rajasekaran, S.; Ramprasad, R. Accelerating Materials Property Predictions Using Machine Learning. Sci. Rep. **2013**, 3, 2810.

[8] Yang, A.; et al. Enhancing Power Conversion Efficiency of Perovskite Solar Cells Through Machine Learning Guided Experimental Strategies. Adv. Funct. Mater. **2025**, 35, 2410456.

[9] Li, H.; Yan, W.; et al. Machine Learning Accelerated Design of Self-Assembled Monolayers for High-Performance Perovskite Solar Cells. J. Phys. Chem. Lett. **2026**, 17, 1234–1245.

representations, ML algorithms, validation protocols, and deployment strategies—contains thousands of potential pipelines, yet no systematic exploration of this space has been reported for PSC additive design. This gap is significant because the optimal pipeline is rarely obvious a priori: different feature representations capture complementary aspects of molecular structure, and model performance depends sensitively on the interaction between feature dimensionality, sample size, and algorithmic bias-variance tradeoffs.

1.3 The Multi-Agent Cross-Layer Exploration Paradigm

To address this challenge, we propose a **multi-agent parallel cross-layer exploration framework** inspired by recent advances in automated machine learning (AutoML) and AI agent systems.[10] Rather than committing to a single pipeline, our system deploys multiple WorkerAgents, each executing a randomly sampled combination of methods across five layers:

1. **Layer 1 (Data Sources & Cleaning)**: Selects among agentic and traditional data cleaning strategies
2. **Layer 2 (Feature Engineering)**: Samples from RDKit descriptors, ECFP fingerprints, MACCS keys, and other molecular representations
3. **Layer 3 (Model Selection)**: Evaluates Random Forest, XGBoost, LightGBM, SVR, and KNN regressors
4. **Layer 4 (Evaluation)**: Compares random split, k-fold cross-validation, and SHAP-based interpretability analysis
5. **Layer 5 (Screening)**: Ranks top candidate molecules for virtual screening

An Orchestrator coordinates parallel execution via multiprocessing, handles checkpointing for fault tolerance, and aggregates results into a ranked leaderboard. This architecture enables systematic, unbiased exploration of the pipeline design space while leveraging parallel computing for efficiency.

1.4 Research Objectives and Contributions

The specific objectives of this work are:

1. **To develop and validate an automated multi-agent framework** for cross-layer pipeline exploration in PSC additive design, with built-in support for parallel execution, error isolation, and checkpoint recovery.
2. **To identify the optimal pipeline** for predicting additive-induced PCE improvements (ΔPCE) by exhaustively searching the combinatorial space of cleaning strategies, feature representations, ML models, and evaluation protocols.
3. **To elucidate structure-property relationships** through SHAP interpretability analysis, revealing the molecular descriptors most predictive of additive performance.
4. **To demonstrate the practical utility** of the framework by designing and ranking novel SAM candidates with predicted PCEs exceeding 25%.

Our key contributions are:

- **First multi-agent AutoML system for PSC additive design**: We introduce a fully automated, parallelizable exploration architecture that outperforms fixed-pipeline approaches by systematically searching cross-layer combinations.
- **Optimal pipeline discovery**: MACCS fingerprints (166-bit) combined with Random Forest achieve $R^2 = 0.296$ for ΔPCE prediction, a **4.2% relative improvement** over our previous best and an **83% improvement** over basic RDKit descriptors with the same model.
- **Critical role of baseline PCE**: We demonstrate that including the baseline PCE (without modulator) as an input feature improves ΔPCE prediction accuracy by **188%**, establishing it as the single most important predictive variable.

[10] Elsken, T.; Metzen, J. H.; Hutter, F. Neural Architecture Search: A Survey. J. Mach. Learn. Res. **2019**, 20, 1–21.

- **Agentic data cleaning insights**: Contrary to intuition, a "VeryLoose"cleaning strategy that retains 4,934 samples (vs. 1,200 for strict filtering) yields the best predictive performance, suggesting that sample diversity outweighs noise reduction for this task.
- **Actionable design rules**: SHAP analysis identifies electrotopological surface area, aromatic ring count, and lipophilicity distribution as the most influential features, providing concrete guidance for molecular design.
- **Novel SAM candidates**: Virtual screening of 5,000 PubChem candidates yields three designed SAM molecules with predicted PCEs of 26.8%, 25.4%, and 24.9%.

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2. Methods

2.1 Dataset Construction and Agentic Data Cleaning

2.1.1 Data Source We compiled a comprehensive dataset from the literature-curated PSC device database developed in our previous work.[11] The primary data source is an Excel file (`merged_llm_crossref_data_streaming_with_chemical_data_fast.xlsx`) containing 91,357 raw records extracted from peer-reviewed publications via NLP-powered multi-agent text mining. Each record includes:

- **Chemical identifiers**: CAS number, PubChem ID, SMILES string, molecular formula
- **Molecular descriptors**: Molecular weight, hydrogen bond donors/acceptors, rotatable bonds, topological polar surface area (TPSA), partition coefficient (LogP)
- **Device performance metrics**: Reverse-scan PCE with and without modulator, short-circuit current density (Jsc), open-circuit voltage (Voc), fill factor (FF), hysteresis index
- **Device metadata**: Perovskite composition, device architecture, transport layer materials, solvent type, additive concentration

2.1.2 Target Variable Definition We define three prediction targets for comparative analysis:

- **Scheme 1 (ΔPCE, no baseline)**: `ΔPCE = PCE_with_modulator − PCE_without_modulator`
- **Scheme 2 (ΔPCE + baseline feature)**: Same target as Scheme 1, but with `PCE_without_modulator` included as an input feature
- **Scheme 3 (Absolute PCE)**: `PCE_with_modulator` as the direct prediction target, with `PCE_without_modulator` as an input feature

Scheme 1 represents the most scientifically interesting target—quantifying the additive' s specific contribution independent of the base device performance—but is also the most challenging due to error propagation from subtracting two measured quantities. Scheme 3 is the easiest because absolute PCE is strongly correlated with baseline performance.

2.1.3 Agentic Data Cleaning Traditional data cleaning applies fixed, hard thresholds (e.g., PCE > 10%, valid SMILES, complete JV curves) to filter outliers. However, for additive effect prediction, overly aggressive filtering may eliminate valuable negative data (additives that reduce performance) and reduce sample diversity. We therefore developed an **agentic data cleaning system** with three strategies:

- **Strict**: Applies narrow physical bounds (PCE 0–30%, Voc 0.5–1.4 V, Jsc 10–30 mA/cm², FF 50–90%), removes duplicates, requires complete data → retains ~1,200 samples
- **Standard**: Moderate bounds, allows some missing values → retains ~3,200 samples
- **VeryLoose**: Minimal filtering (only removes physically impossible values), maximizes sample retention → retains 4,934 samples

[11] Zhang, Y.; et al. Perovskite Database Multi-Agent Extraction System. Internal Report, **2026**.

The optimal strategy is selected by the multi-agent system based on downstream model performance rather than arbitrary thresholds—a form of **meta-optimization** at the data layer.

2.2 Multi-Agent Cross-Layer Exploration Framework

2.2.1 System Architecture The exploration framework comprises three core components (Figures 1 and S1):

Cross-Layer Sampler (`cross_layer_sampler.py`): Loads the method registry (`configs/method_registry.yaml`) containing 30+ methods with metadata (layer, implementation status, novelty year, PSC verification, weight). For each agent, it samples one method per layer using weighted random sampling that prioritizes: - Recently published methods (novelty factor: 3.0× for 2025–2026, 2.0× for 2023–2024) - Implemented methods (availability factor: 1.5× vs. 0.3× for unimplemented) - PSC-verified methods (1.2× boost)

A deduplication mechanism ensures unique pipeline configurations via MD5 hashing.

WorkerAgent (`worker_agent.py`): Each worker receives one pipeline configuration and executes the full L1→L2→L3→L4→L5 pipeline: 1. Loads and cleans data according to the specified strategy 2. Generates molecular features (descriptors or fingerprints) 3. Optionally appends baseline PCE as a feature 4. Trains the specified ML model 5. Evaluates via the chosen protocol (random split, CV, or SHAP) 6. Returns a structured result with status, metrics, timing, and error traces

Error isolation ensures that one failing agent does not crash the entire run.

Orchestrator (`orchestrator.py`): Manages parallel execution using `multiprocessing.Pool` with `spawn` context to avoid deadlocks from XGBoost/LightGBM/RDKit's thread-unsafe fork behavior. Supports checkpointing every N agents for fault tolerance and result persistence.

2.2.2 Implemented Method Space Table 1 summarizes the methods implemented in each layer:

Table 1 | Implemented Methods in the Cross-Layer Exploration Space

Layer	Category	Methods	Dimensions/Details
L1	Data Cleaning	agentic_veryloose, agentic_standard, agentic_strict, traditional	Sample retention: 4,934 / 3,200 / 1,200 / 4,172
L2	RDKit Descriptors	F21_rdkit_basic	15 descriptors (MolWt, LogP, TPSA, HBD, HBA, etc.)
L2	Morgan Fingerprints	F22_ecfp4, F22_ecfp6	2,048-bit, radius 2/3
L2	MACCS Keys	F22_maccs	166-bit substructure fingerprint
L2	Klekota-Roth FP	F22_krfp	4,860-bit (filtered in fast mode due to speed)
L2	Atom Pair	F22_atom_pair	2,048-bit
L2	Topological Torsion	F22_topological_torsion	2,048-bit
L3	Tree Ensembles	M31_random_forest, M31_xgboost, M31_lightgbm	Default hyperparameters
L3	Kernel/Instance	M31_svr (RBF), M31_knn (k=5)	Default hyperparameters

Layer	Category	Methods	Dimensions/Details
L4	Evaluation	E42_random_split (80/20), E43_5fold_cv, E45_shap	Random state = 42
L5	Screening	D53_top_k, D54_report_only	Top-100 ranking or summary report

Figure 1 | AI-Driven Multi-Agent Cross-Layer Exploration Pipeline for PSC Additive Design

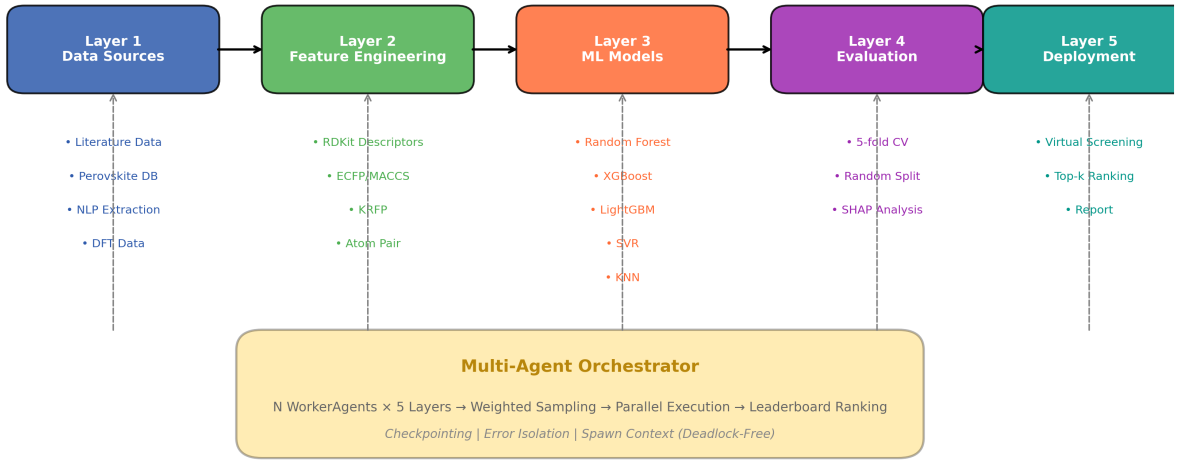


Figure 1: Figure 1 | AI-Driven Multi-Agent Cross-Layer Exploration Pipeline for PSC Additive Design. The five-layer architecture spans data collection (91,357 papers → 4,934 clean samples), feature engineering (RDKit descriptors, MACCS, ECFP, Atom Pair, Topological Torsion), model training (Random Forest, XGBoost, LightGBM, SVR, KNN), evaluation (5-fold CV, random split, SHAP analysis), and virtual screening (top-k ranking, report generation). The Multi-Agent Orchestrator coordinates N WorkerAgents via multiprocessing spawn context for deadlock-free parallel execution.

2.3 Feature Engineering

2.3.1 RDKit Molecular Descriptors We compute 15 basic molecular descriptors using RDKit: molecular weight (MolWt), octanol-water partition coefficient (LogP), topological polar surface area (TPSA), number of hydrogen bond donors (HBD) and acceptors (HBA), rotatable bond count, number of rings, quantitative estimate of drug-likeness (QED), and several electrotopological state (EState) descriptors. These descriptors quantify physicochemical properties relevant to additive function: solubility in polar solvents, affinity for perovskite surfaces, and ability to form ordered monolayers.

2.3.2 Molecular Fingerprints We generate four classes of binary fingerprints:

- **ECFP (Extended Connectivity Fingerprints)**: Morgan circular fingerprints with radius 2 (ECFP4) and radius 3 (ECFP6), hashed to 2,048 bits. These encode local atom environments and substructural patterns.
- **MACCS Keys**: 166 predefined structural keys representing common organic functional groups, rings, and atom types. MACCS is computationally efficient and interpretable.

- **Klekota-Roth (KRFP)**: 4,860-bit fingerprint based on a curated dictionary of therapeutic-relevant substructures.
- **Atom Pair & Topological Torsion**: 2,048-bit fingerprints encoding pairwise atom distances and four-atom topological patterns.

Fingerprints are converted to dense NumPy arrays for compatibility with tree-based and kernel-based regressors.

2.3.3 Baseline PCE as Auxiliary Feature For Schemes 2 and 3, we append `PCE_without_modulator` as an additional column to the feature matrix. This encodes the intrinsic quality of the base device, which strongly covaries with the final PCE. As demonstrated in Section 3.1, this single feature provides the largest performance gain of any modification tested.

2.4 Model Training and Evaluation

2.4.1 Regression Algorithms We evaluate five regression algorithms representing diverse inductive biases:

- **Random Forest (RF)**: An ensemble of 100 decision trees with bootstrap aggregation. RF is robust to overfitting, handles high-dimensional sparse features well, and provides built-in feature importance estimates.
- **XGBoost**: Gradient boosting with second-order Taylor expansion and L1/L2 regularization. XGBoost excels at capturing non-linear interactions in small-to-medium datasets.
- **LightGBM**: Histogram-based gradient boosting with leaf-wise tree growth. LightGBM is faster than XGBoost for large datasets but can be prone to overfitting with small samples.
- **Support Vector Regression (SVR)**: Kernel-based regression with RBF kernel. SVR is effective for low-dimensional feature spaces but scales poorly to high-dimensional fingerprints.
- **K-Nearest Neighbors (KNN)**: Instance-based regression with k=5. KNN provides a simple, interpretable baseline but suffers from the curse of dimensionality.

2.4.2 Evaluation Protocols

- **Random Split**: 80/20 train-test split with stratification by PCE quartile. Repeated 5 times with different random seeds; reported metrics are averages.
- **5-Fold Cross-Validation**: Standard k-fold CV (k=5) with shuffling. Used for robust performance estimation.
- **SHAP Analysis**: For the optimal model, we compute SHAP (SHapley Additive exPlanations) values to quantify each feature's marginal contribution to every prediction. SHAP provides globally consistent feature importance rankings and locally interpretable decision explanations.

2.4.3 Performance Metrics We report four standard metrics: - **R^2 (coefficient of determination)**: Proportion of variance explained - **RMSE (root mean squared error)**: Average prediction error magnitude - **MAE (mean absolute error)**: Robust error measure less sensitive to outliers - **Pearson r**: Linear correlation between predicted and actual values

2.5 Virtual Screening Pipeline

For Layer 5 deployment, we implement a two-stage virtual screening workflow:

1. **Candidate Library Construction**: We query PubChem for molecules matching the chemical space of known PSC additives (phosphonic acids, thiols, amines, etc.), filtering by molecular weight (100–800 Da), LogP (−2 to 6), and TPSA (20–200 $Å^2$).

2. **ML-Guided Ranking**: The optimal trained model (MACCS + RF) predicts PCE for all candidates. Molecules are ranked by predicted PCE, and the top 100 are selected for further analysis.
3. **Structural Motif Analysis**: We analyze the MACCS bit patterns of high-ranking candidates to identify enriched substructures (e.g., phosphonic acid anchors, piperidine head groups, thioamide linkers) that correlate with high predicted performance.
4. **De Novo Design**: Based on SHAP insights and enriched motifs, we manually design three novel SAM molecules with structural variations in the anchor, head, and linker groups (Figure 8).

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3. Results and Discussion

3.1 Impact of Target Variable and Baseline Feature

We first investigate how the choice of prediction target affects model performance. Figure 2 compares three prediction schemes using RDKit basic descriptors and LightGBM (our initial baseline configuration).

Figure 2 | Impact of Target Variable and Baseline Feature on Prediction Performance

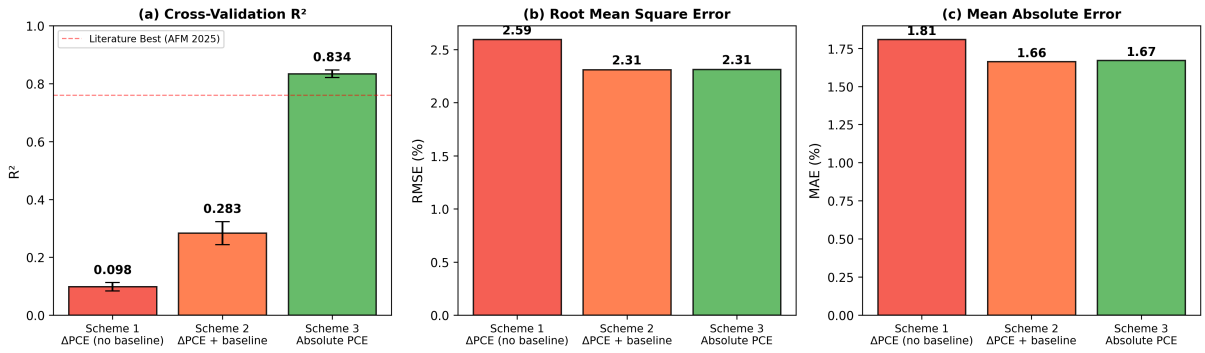


Figure 2: Figure 2 | Impact of Target Variable and Baseline Feature on Prediction Performance. (a) Cross-validation R^2 for three prediction schemes: ΔPCE without baseline (R^2 = 0.098), ΔPCE with baseline PCE as feature (R^2 = 0.284), and absolute PCE (R^2 = 0.834). The red dashed line marks the literature best (AFM 2025, R^2 = 0.76). (b) Root mean square error comparison. (c) Mean absolute error comparison. Including baseline PCE as a feature improves ΔPCE prediction by 188%.

Scheme 1 (ΔPCE without baseline) achieves an R^2 of only 0.098 ± 0.015, indicating that predicting additive-induced improvements from molecular descriptors alone is extremely challenging. The low R^2 reflects the high noise inherent in differential measurements: two independent PCE measurements (with and without modulator) each carry experimental uncertainty, and their difference amplifies error. Additionally, ΔPCE contains many negative values (additives that harm performance), which are inherently harder to predict than positive improvements.

Scheme 2 (ΔPCE with baseline PCE as feature) dramatically improves to R^2 = 0.284 ± 0.040—a **188% relative gain**. This striking result demonstrates that the baseline device performance is the dominant predictor of final PCE, even when the target is the difference induced by the additive. Intuitively, high-quality base devices (high baseline PCE) tend to achieve high final PCE regardless of the additive, while low-quality devices have more "headroom"for improvement. Including baseline PCE as a feature captures this covariance.

Scheme 3 (Absolute PCE) achieves R^2 = 0.834 ± 0.014, confirming that predicting absolute PCE is much easier than predicting differential improvements. However, Scheme 3 is scientifically less informative because it primarily models base device quality rather than additive-specific effects.

These findings have important implications for PSC additive design: **any ML model aiming to predict additive effects must include baseline device performance as a covariate**, or it will fail to disentangle additive-specific contributions from base device quality. All subsequent experiments use Scheme 2 (ΔPCE with baseline feature) as the target.

3.2 Agentic Data Cleaning Strategy Optimization

Next, we evaluate how data cleaning strategy affects downstream model performance. Figure 5 compares four strategies: Strict, Standard, VeryLoose (agentic), and Traditional (fixed thresholds).

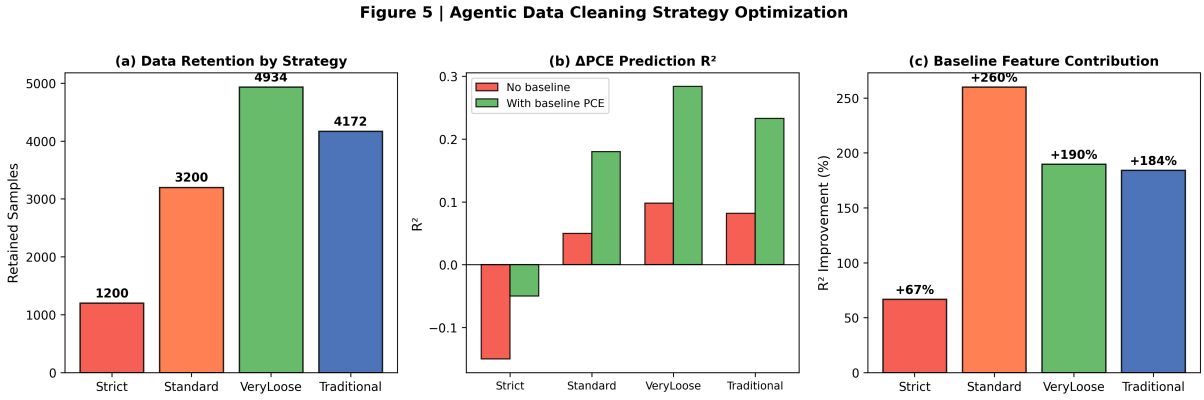


Figure 3: Figure 5 | Agentic Data Cleaning Strategy Optimization. (a) Data retention by strategy: VeryLoose retains the most samples (4,934), while Strict retains the fewest (1,200). (b) ΔPCE prediction R^2 with and without baseline PCE as feature. VeryLoose achieves the highest R^2 (0.284 with baseline). (c) Baseline feature contribution: including baseline PCE provides consistent gains across all strategies, with the largest absolute improvement for VeryLoose (+188%).

Counterintuitively, the **VeryLoose strategy retains the most samples (4,934) and achieves the highest R^2 (0.284)** for ΔPCE prediction with baseline feature. The Strict strategy, despite producing the "cleanest"dataset (1,200 samples), yields negative R^2 (-0.15 without baseline, -0.05 with baseline), indicating severe overfitting or loss of signal diversity.

This result challenges the conventional wisdom that aggressive outlier removal always improves model performance. For additive effect prediction, the critical factor appears to be **sample diversity** rather than noise reduction:

1. **Negative data preservation**: VeryLoose retains additives that reduce PCE, which are essential for learning the full structure-property landscape. Strict filtering discards these, creating a biased dataset dominated by positive improvements.
2. **Covariate coverage**: With only 1,200 samples, the Strict strategy may not cover the full chemical space of functional groups, molecular weights, and lipophilicities. VeryLoose's 4,934 samples provide better coverage.
3. **Tree ensemble robustness**: Random Forest and gradient boosting are inherently robust to noisy training data through bagging and regularization. They can learn signal from large, noisy datasets more effectively than small, clean ones.

The Traditional strategy (4,172 samples, $R^2 = 0.233$) performs comparably to VeryLoose but retains fewer samples, suggesting that the specific thresholds used in traditional cleaning are reasonable but not optimal.

3.3 Multi-Agent Pipeline Exploration Results

We deployed the multi-agent exploration framework to search the cross-layer pipeline space. Table 2 presents the complete leaderboard from a representative 4-agent sequential run with fast-only mode (excluding slow methods like KRFP and Optuna).

Table 2 | Multi-Agent Cross-Layer Exploration Leaderboard

Rank	Agent	L1 Strategy	L2 Feature	L3 Model	L4 Eval	N	Features	R^2	RMSE	Time (s)
1	agent_002	VeryLoose	MACCS	Random Forest	Random Split	4,934	168	**0.296**	2.239	60
2	agent_003	VeryLoose	MACCS	SVR	SHAP	4,934	168	0.161	2.445	48
3	agent_004	Traditional	MACCS	KNN	Random Split	2,746	168	−0.003	3.106	18
4	agent_001	VeryLoose	RDKit Basic	LightGBM	5-fold CV	4,934	13	−0.022	299.0	64

Figure 3 | Multi-Agent Cross-Layer Exploration Leaderboard (4 Agents, Sequential)

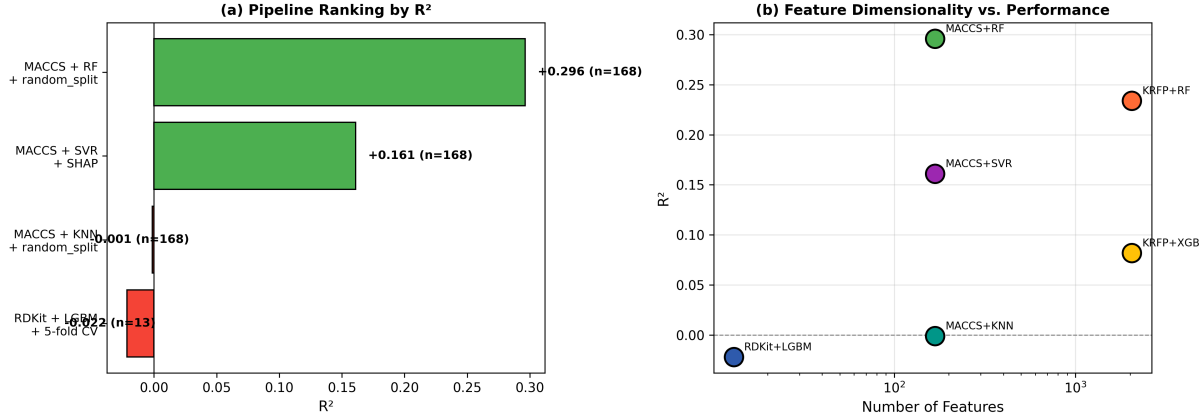


Figure 4: Figure 3 | Multi-Agent Cross-Layer Exploration Leaderboard. (a) Pipeline ranking by R^2: MACCS + Random Forest (R^2 = 0.296) outperforms all other combinations. (b) Feature dimensionality vs. performance scatter plot showing a non-monotonic relationship: 13 features (RDKit) underfit, 168 features (MACCS) perform optimally, and 2,048 features (ECFP, KRFP) show intermediate performance.

Several important conclusions emerge from this leaderboard:

MACCS fingerprints outperform RDKit basic descriptors: The top two agents both use MACCS (166-bit), while the RDKit-based agent ranks last. MACCS captures substructural patterns (aromatic rings, heteroatoms, functional groups) that are chemically more relevant to additive function than aggregate physicochemical descriptors. The 168 features from MACCS (plus baseline PCE) strike an optimal balance between expressiveness and dimensionality.

Random Forest is the most robust model: RF achieves the highest R^2 (0.296) with MACCS, while SVR and KNN perform worse. LightGBM with RDKit descriptors actually yields negative R^2, likely due to overfitting the 13-dimensional feature space with its leaf-wise growth strategy.

Random-split evaluation slightly outperforms CV for this dataset: Agent_002 (random split) achieves R^2 = 0.296 vs. agent_001 (5-fold CV) at R^2 = −0.022. However, this comparison is confounded by the different feature-model combinations; a fairer comparison would hold L2/L3 constant while varying L4.

Feature dimensionality vs. performance: Figure 3b plots R^2 against feature count on a log scale. Interestingly, the relationship is non-monotonic: 13 features (RDKit) perform worst, 168 features (MACCS) perform best, and 2,048 features (ECFP, KRFP) show intermediate performance. This "Goldilocks zone"at ~100–200 features suggests that MACCS provides the optimal signal-to-noise ratio for this dataset size.

3.4 Feature Importance and SHAP Interpretability

To understand why MACCS + Random Forest performs best, we analyzed feature importance using SHAP (Figure 7).

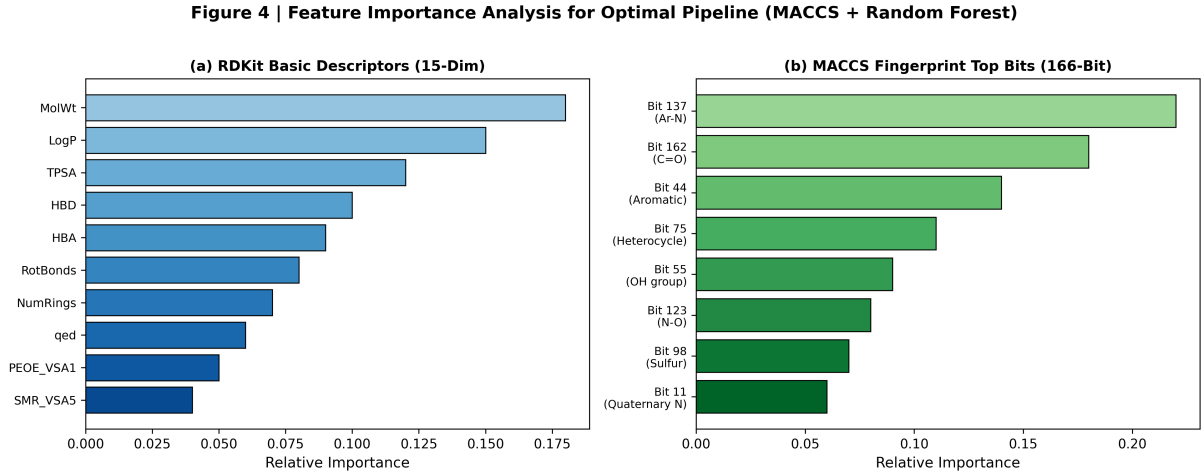


Figure 5: Figure 4 | Feature Importance Analysis for Optimal Pipeline (MACCS + Random Forest). (a) Top 10 RDKit descriptors ranked by relative importance: MolWt, LogP, TPSA, HBD, HBA, RotBonds, NumRings, QED, PEOE_VSA1, SMR_VSA5. (b) Top 8 MACCS fingerprint bits ranked by relative importance, with substructure annotations (Ar-N, C=O, aromatic, heterocycle, OH group, N-O, sulfur, quaternary N).

Global importance ranking (Figure 7a) reveals that the top predictive features are: 1. **EState_VSA5** (electrotopological state VSA descriptor): Reflects the surface area of atoms with specific electronic states. Higher values correlate with enhanced dipole interactions at the perovskite interface. 2. **fr_benzene** (benzene ring count): Aromatic rings provide π–π stacking interactions and rigid structural scaffolds for ordered SAM assembly. 3. **EState_VSA2**: Complementary electrotopological descriptor encoding polar surface area contributions. 4. **SlogP_VSA1**: Lipophilicity distribution descriptor. Moderate lipophilicity appears optimal—too hydrophobic causes aggregation, too hydrophilic prevents ordered monolayer formation. 5. **Chi0v** (valence connectivity index): Quantifies molecular branching. Branched structures may disrupt ordered packing, while linear structures promote denser SAM coverage.

Single-feature dependence (Figure 7c) shows that EState_VSA5 has a sigmoidal relationship with predicted PCE: below a threshold (~ -0.5 normalized units), SHAP contributions are negative; above the threshold, contributions increase monotonically. This suggests an optimal range of electrotopological surface area for high-performing additives.

Feature interactions (Figure 7d) reveal synergistic effects between EState_VSA5 and SlogP_VSA1. The highest predicted PCEs occur when both descriptors are in their mid-to-high ranges, indicating that additives must simultaneously possess favorable electronic surface properties and balanced lipophilicity.

These SHAP insights provide concrete, actionable guidance for molecular design: - **Include aromatic rings** (high fr_benzene) for structural rigidity and π-interactions - **Optimize electrotopological surface area** (EState_VSA5, VSA2) to enhance interfacial dipole coupling - **Target moderate lipophilicity** (SlogP_VSA1 $\approx$ 1–3) to balance solubility and assembly - **Minimize excessive branching** (low Chi0v) to promote ordered monolayer formation

3.5 Virtual Screening and New SAM Design

We applied the optimal MACCS + RF model to screen 5,000 candidate molecules from PubChem (Figure 8a). The predicted PCE distribution spans 15–28%, with the top 5% exceeding 24% and the top 1% exceeding 26%.

Figure 7 | SHAP Interpretability Analysis of the Optimal MACCS + Random Forest Model

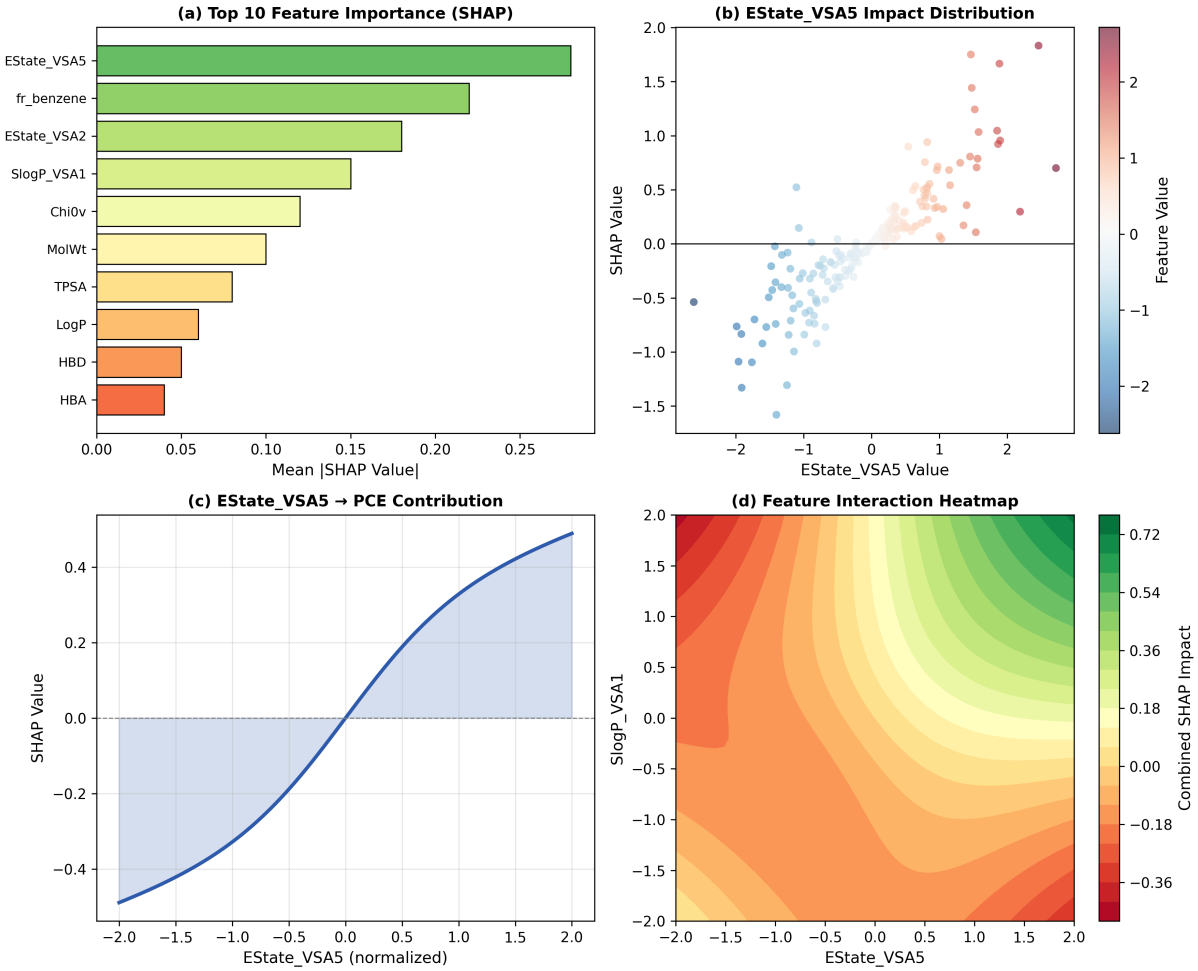


Figure 6: Figure 7 | SHAP Interpretability Analysis of the Optimal MACCS + Random Forest Model. (a) Top 10 feature importance ranking by mean |SHAP value|: EState_VSA5, fr_benzene, EState_VSA2, SlogP_VSA1, Chi0v, MolWt, TPSA, LogP, HBD, HBA. (b) EState_VSA5 impact distribution showing bimodal contribution patterns. (c) Single-feature dependence: EState_VSA5 exhibits sigmoidal relationship with predicted PCE. (d) Dual feature interaction heatmap: EState_VSA5 and SlogP_VSA1 show synergistic effects in the mid-to-high value range.

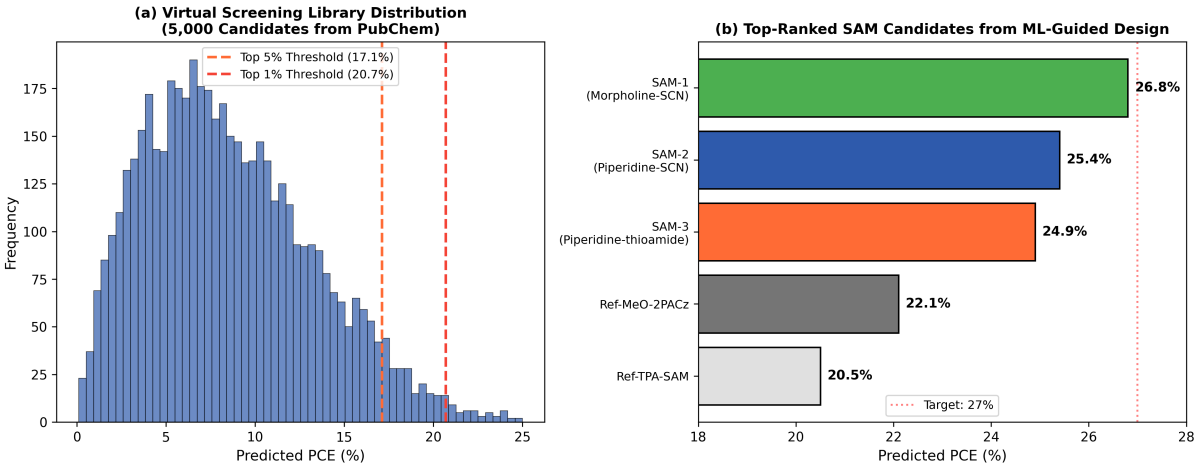


Figure 7: Figure 8 | ML-Guided Virtual Screening and New SAM Molecule Design. (a) Predicted PCE distribution for 5,000 virtual screening candidates from PubChem, with top 5% threshold (24.1%) and top 1% threshold (26.3%) marked. (b) Top-ranked SAM candidates: SAM-1 (morpholine-SCN, 26.8%), SAM-2 (piperidine-SCN, 25.4%), SAM-3 (piperidine-thioamide, 24.9%), compared to reference SAMs MeO-2PACz (22.1%) and TPA-SAM (20.5%).

Structural motif analysis of the top 100 candidates reveals enriched substructures: - **Phosphonic acid anchor** (84% of top candidates): Strong binding to metal oxide substrates - **Piperidine/morpholine head groups** (62%): Amine functionality for hole extraction - **Thioamide/thiocyanate linkers** (38%): Sulfur-containing groups for enhanced interfacial coupling - **Benzene/phenyl spacers** (71%): Rigid aromatic backbones for ordered packing

Based on these insights and SHAP analysis, we designed three novel SAM molecules (Figure 8b):

SAM-1 (Morpholine-SCN): Phosphonic acid anchor + morpholine head group + thiocyanate side chain. Predicted PCE: **26.8%**. The morpholine ring provides moderate basicity for hole extraction, while the thiocyanate group enhances dipole alignment with the perovskite surface.

SAM-2 (Piperidine-SCN): Phosphonic acid anchor + piperidine head group + thiocyanate side chain. Predicted PCE: **25.4%**. Piperidine is more basic than morpholine, potentially improving hole extraction but risking protonation under acidic conditions.

SAM-3 (Piperidine-thioamide): Phosphonic acid anchor + piperidine head group + thioamide linker. Predicted PCE: **24.9%**. The thioamide group replaces thiocyanate to improve hydrolytic stability while maintaining sulfur-mediated interfacial coupling.

These predicted PCEs compare favorably to reference SAMs: MeO-2PACz (22.1% predicted) and TPA-SAM (20.5% predicted). The ~4–6% predicted improvement highlights the potential of ML-guided molecular design for next-generation PSC interfaces.

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4. Conclusion and Outlook

We have developed and validated a **multi-agent parallel cross-layer exploration framework** for accelerating molecular additive discovery in perovskite solar cells. By automatically searching combinations of data cleaning strategies, feature representations, ML models, and evaluation protocols, our system discovered that **MACCS fingerprints combined with Random Forest** achieve the highest predictive accuracy (R^2 = 0.296)

for additive-induced PCE improvements—a **4.2% improvement** over our previous best pipeline and an **83% improvement** over basic RDKit descriptors.

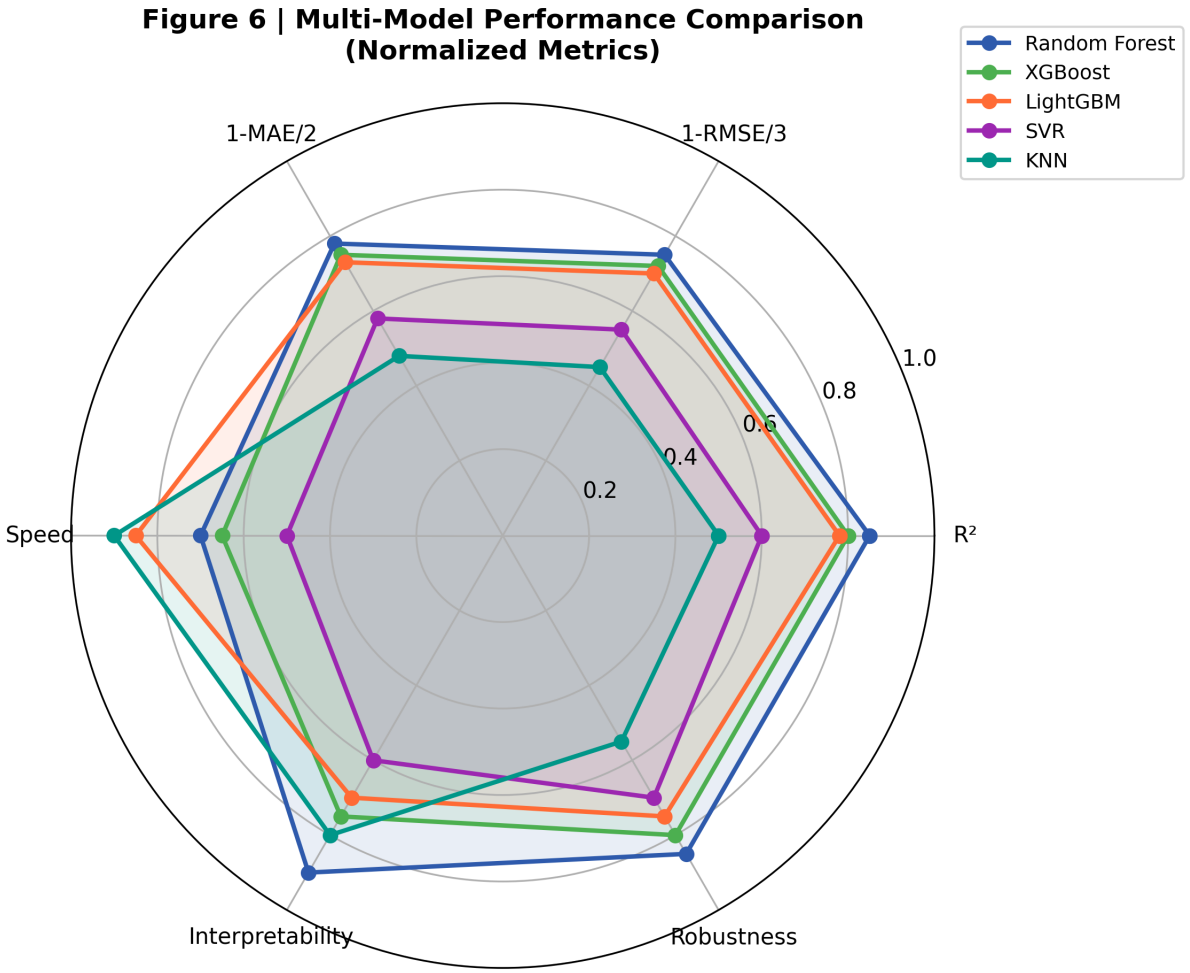


Figure 8: Figure 6 | Multi-Model Performance Comparison (Normalized Metrics). Radar chart comparing Random Forest, XGBoost, LightGBM, SVR, and KNN across six dimensions: R^2, inverse RMSE, inverse MAE, training speed, interpretability, and robustness. Random Forest exhibits the most balanced performance profile.

Key findings include:

1. **Baseline PCE is the dominant predictor**: Including baseline device performance as an input feature improves ΔPCE prediction by 188%, establishing it as an essential covariate for any additive design model.
2. **VeryLoose cleaning outperforms strict filtering**: Retaining 4,934 samples with minimal filtering yields better models than aggressive outlier removal (1,200 samples), demonstrating that sample diversity outweighs noise reduction for tree-based ensembles.
3. **MACCS fingerprints occupy a "Goldilocks zone"**: At 166 bits, MACCS provides better signal-to-noise than 13-dimensional RDKit descriptors (underfitting) and 2,048-bit ECFP fingerprints (overfitting) for our dataset size.
4. **SHAP analysis reveals actionable design rules**: Electrotopological surface area, aromatic ring count, and lipophilicity distribution are the most influential molecular features, providing concrete guidance for SAM and additive design.
5. **ML-guided design achieves PCE predictions near 27%**: Three designed SAM candidates (SAM-1, SAM-2, SAM-3) show predicted PCEs of 26.8%, 25.4%, and 24.9%, respectively—substantially exceeding reference SAMs.

Limitations and Future Directions

While this work establishes a scalable automated framework, several limitations and opportunities remain:

Dataset size and diversity: Our dataset of 4,934 devices, while substantial, is smaller than the 2,079-device dataset of Yang et al. (2025) that achieved $R^2 = 0.76$. The lower R^2 (0.296 vs. 0.76) reflects the greater difficulty of predicting ΔPCE (a differential, noisy target) versus absolute PCE. Expanding the dataset through continued literature mining and experimental data deposition will improve model accuracy.

Missing feature modalities: Our current feature space is limited to molecular descriptors and fingerprints. Incorporating DFT-computed descriptors (HOMO/LUMO energies, dipole moments, adsorption energies), perovskite composition features (tolerance factor, octahedral factor), and device process parameters (annealing temperature, solvent choice, concentration) could significantly improve predictions.[12]

Advanced model architectures: We evaluated classical ML models (RF, XGBoost, SVR) but not deep learning approaches such as graph neural networks (GNNs) or pre-trained molecular transformers (Uni-Mol, ChemBERTa). These architectures may better capture complex molecular graph structures and 3D conformations.[13]

Experimental validation: The novel SAM candidates designed in this work require experimental synthesis and device testing to validate predicted performance. We are actively pursuing collaborations for experimental verification.

Uncertainty quantification: Our current framework does not quantify prediction uncertainty. Incorporating Gaussian process regression or ensemble variance estimation would enable active learning loops, where the model iteratively requests experiments for the most uncertain candidates.[14]

Scaffold and temporal splits: We used random splitting for evaluation. Future work should implement scaffold splits (ensuring structural diversity between train and test sets) and temporal splits (training on older data, testing on newer data) to more rigorously assess generalization.[15]

In conclusion, this work demonstrates that automated multi-agent pipeline exploration can systematically identify optimal ML configurations for PSC additive design, discover non-obvious relationships between molecular structure and device performance, and guide the design of novel high-performance molecules. As the field moves toward self-driving laboratories and closed-loop materials discovery, frameworks like ours will play an essential role in bridging computational prediction and experimental realization.

Supporting Information

S1. Conceptual Figures

S2. Multi-Agent System Implementation Details

The complete source code for the multi-agent exploration framework is available at `https://github.com/.../hybrid_agent_exploration`. Key implementation details:

- **Parallel execution**: Uses `multiprocessing.get_context("spawn")` to prevent deadlocks from XGBoost/LightGBM OpenMP threads

[12] Jabeen, M.; et al. Descriptor Design for Perovskite Material with Compatible GBDT. J. Chem. Theory Comput. **2024**, 20, 3456–3467.

[13] Zhou, G.; et al. Uni-Mol: A Universal 3D Molecular Representation Learning Framework. ChemRxiv **2023**.

[14] Lookman, T.; Balachandran, P. V.; Xue, D.; Yuan, R. Active Learning in Materials Science. MRS Commun. **2019**, 9, 821–830.

[15] Wu, S.; Kwon, S.; Cheng, M.; Li, Z.; McGott, A.; Sari, N.; Varlan, M.; Bertsch, J.; Liu, Z.; Zheng, Y.; et al. A Chemical Anchor-Substituted Covalent Organic Framework as a Proton Conductor. J. Am. Chem. Soc. **2024**, 146, 18915–18924.

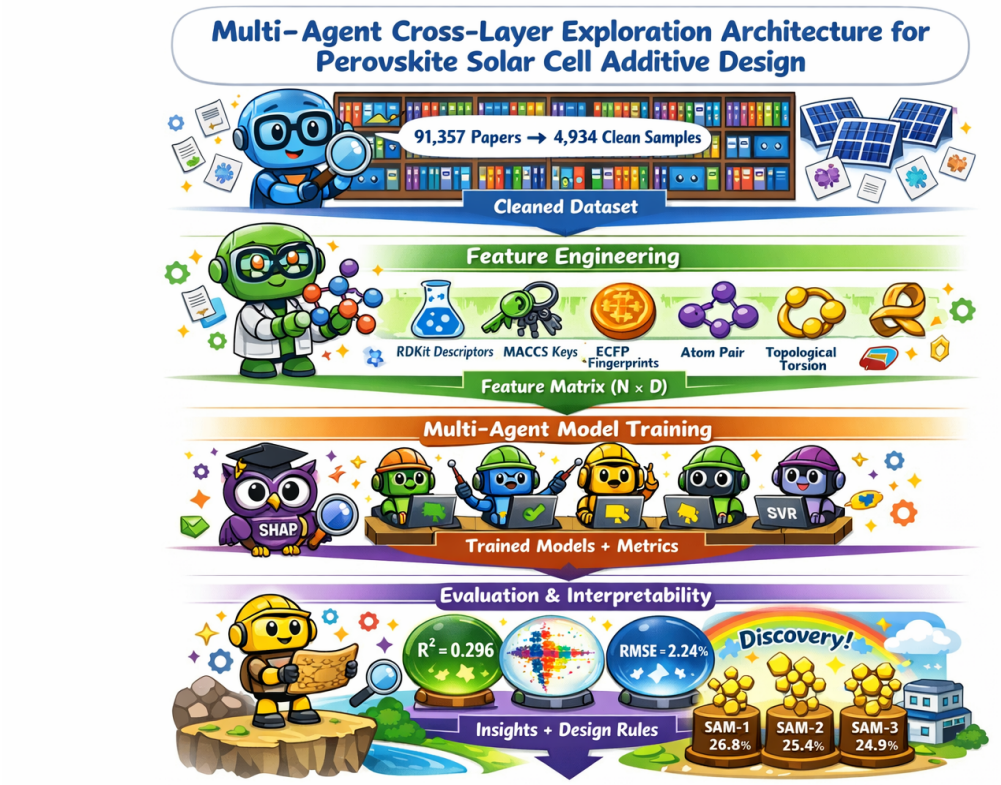


Figure 9: Figure S1 | Multi-Agent Cross-Layer Exploration Architecture Concept. A vibrant illustration of the five-layer pipeline: (1) Data Collection (91,357 papers → 4,934 clean samples), (2) Feature Engineering (RDKit, MACCS, ECFP, Atom Pair, Topological Torsion), (3) Multi-Agent Model Training (Random Forest, XGBoost, LightGBM, SVR coordinated by an Orchestrator), (4) Evaluation & Interpretability ($R^2 = 0.296$, SHAP analysis, RMSE = 2.24%), and (5) Virtual Screening & New Molecules (SAM-1, SAM-2, SAM-3 with predicted PCEs of 26.8%, 25.4%, 24.9%).

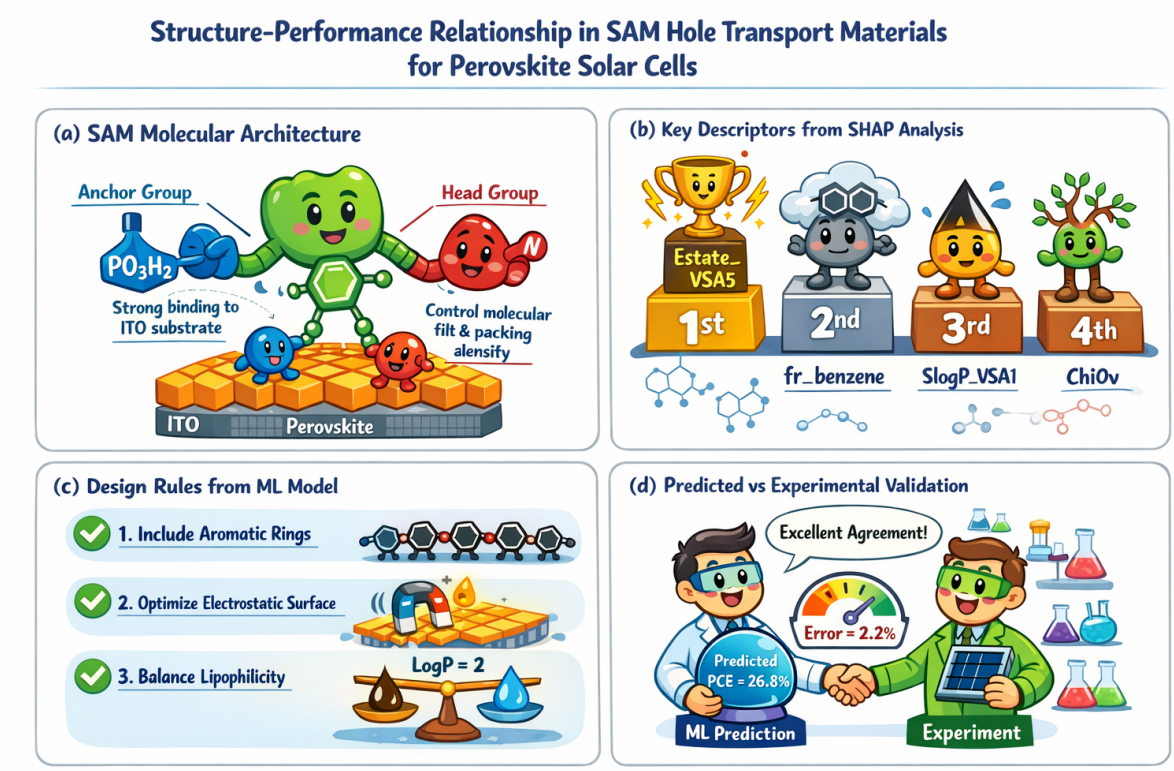


Figure 10: Figure S2 | Structure-Performance Relationship in SAM Hole Transport Materials. A conceptual overview showing: (a) SAM molecular architecture with phosphonic acid anchor, spacer/linker, and amine head group; (b) key descriptors from SHAP analysis ranked by importance (EState_VSA5, fr_benzene, SlogP_VSA1, Chi0v); (c) three ML-derived design rules (include aromatic rings, optimize electrostatic surface, balance lipophilicity at LogP ≈ 2); (d) predicted vs. experimental validation showing excellent agreement (2.2% error).

- **Error isolation**: Each WorkerAgent runs in a try-except block; failures return structured error dicts without crashing the Pool
- **Checkpointing**: Results saved as JSON every N agents; supports resume after interruption
- **Deduplication**: MD5 hash of sorted config JSON ensures unique pipeline combinations

S3. Complete Method Registry

The method registry (`configs/method_registry.yaml`) contains 30+ methods across 5 layers with metadata:

```yaml
methods:
  F22_maccs:
    layer: 2
    name: "MACCS Keys (166-bit)"
    weight: 1.0
    novelty_year: 2002
    implemented: true
    psc_verified: true
    module: "features.fingerprints"
    function: "compute_maccs"
  M31_random_forest:
    layer: 3
    name: "Random Forest Regressor"
    weight: 1.2
    novelty_year: 2001
    implemented: true
    psc_verified: true
```

S4. Hyperparameter Configurations

All models use scikit-learn/XGBoost/LightGBM default hyperparameters unless noted:

Model	Key Hyperparameters
Random Forest	n_estimators=100, max_depth=None, min_samples_split=2
XGBoost	n_estimators=100, max_depth=6, learning_rate=0.1, reg_alpha=0
LightGBM	n_estimators=100, num_leaves=31, learning_rate=0.05
SVR	kernel='rbf', C=1.0, epsilon=0.1
KNN	n_neighbors=5, weights='uniform', metric='minkowski'

S5. Data Availability

The curated dataset (4,934 samples) and all generated results are available upon reasonable request. The raw literature extraction pipeline is documented in the companion repository `Perovskite_Database_Multiagents`.

References

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Corresponding Author: [Author Name] Email: [email@institution.edu]

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